## Querying DocumentDB

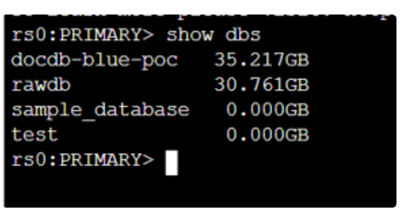
There are various options for querying the DocumentDB.

1. Connect to mongo shell through EC2.
2. Connect through Athena using AWS console.
3. Studio 3T (IDE tool).
4. MongoDB Compass (SSH to EC2 instance).
5. VS Code extension for MongoDB.

### 1. Connect to Mongo shell using EC2

**Prerequisites:** DocumentDB cluster and EC2 instance up and running.

**Reference:** [Connect Amazon EC2 automatically - Amazon DocumentDB]()

1. Go to the EC2 service through AWS management console.
2. Filter on "[i-024e335c9766014bb]()" and open it in lower.
3. Connect through the session manager, EC2 connection should open up.
4. Execute below command for importing the certificate.
    a. sudo wget https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
5. Change directory
    a. cd /u01/app/certs/
6. Run the below command to connect to DocumentDB instance.
    a. mongo --tls --host docdb-blue-poc.cluster-clklrekkgrvy.us-east-1.docdb.amazonaws.com:27017 --tlsCAFile global-bundle.pem --username blueadmin --password <<provide password here>>
7. Execute command **show dbs** for querying the existing databases.
    
    ```
    rs0:PRIMARY> show dbs
    docdb-blue-poc    35.217GB
    rawdb             30.761GB
    sample_database    0.000GB
    test               0.000GB
    rs0:PRIMARY>
    ```
    a.
8. Execute command **use <<DB name>>** to switch to that database.
9. Execute below command to query all the available collections
    a.
    ```
    db.runCommand(
    ```

```
{
listCollections: 1.0,
authorizedCollections: true,
nameOnly: true
}
)

```

```
rs0:PRIMARY> db.runCommand(
... {
... listCollections: 1.0,
... authorizedCollections: true,
... nameOnly: true
... }
... )
{
        "waitedMS" : NumberLong(0),
        "cursor" : {
                "firstBatch" : [
                        {
                                "name" : "example",
                                "type" : "collection"
                        },
                        {
                                "name" : "harikatestcollection",
                                "type" : "collection"
                        },
                        {
                                "name" : "testcoll",
                                "type" : "collection"
                        }
                ],
                "id" : NumberLong(0),
                "ns" : "test.$cmd.listCollections"
        },
        "ok" : 1,
        "operationTime" : Timestamp(1720923127, 1)
}
```

b.

10. Execute **db.harikatestcollection.find( {} ).pretty()** for displaying all records.
11. Execute **db.harikatestcollection.find({"member.memberId": "7485022222" }).pretty()** to filter on specific records.
12. Index creation: **db.runCommand({"createIndexes":"harikatestcollection","indexes":[{"key": {"memberid":1}, "name":"memberid_idx"}]})**

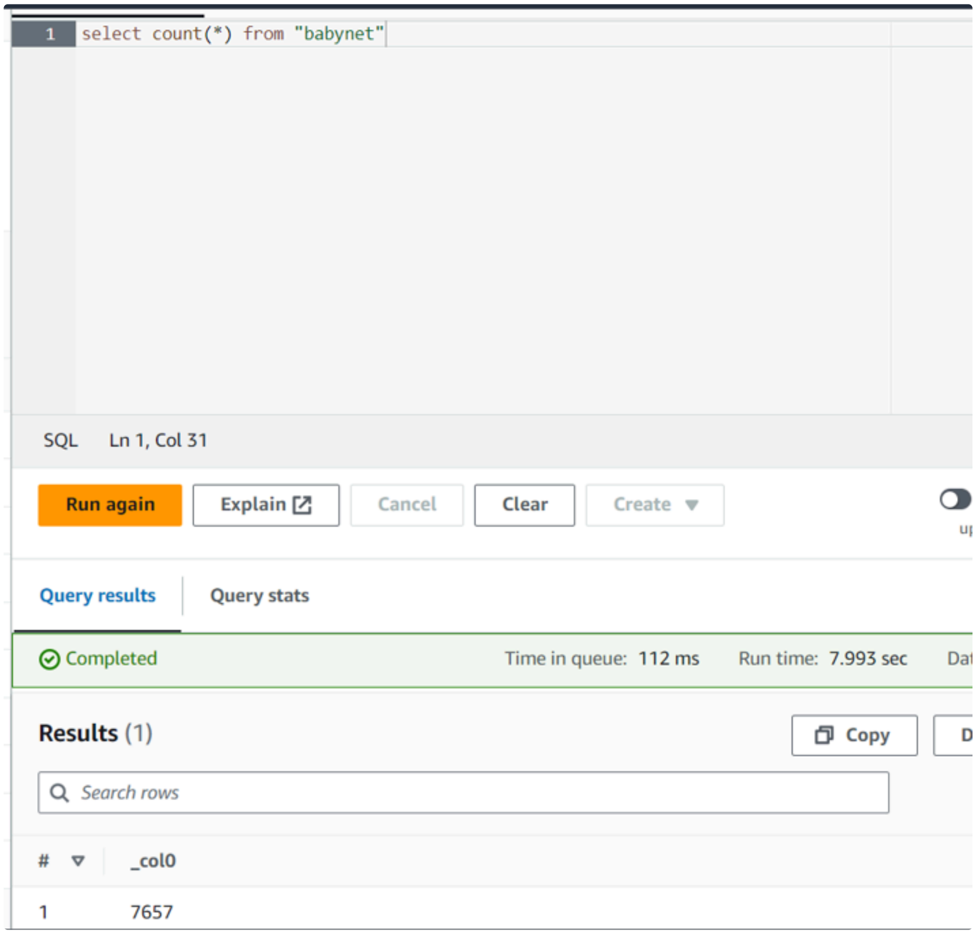
### 2. Connect through Athena using AWS Management console.

**Prerequisites:** DocumentDB cluster and Athena data source should exist.

**Athena data source:** Data source created for this POC is **BlueDocumentDB**

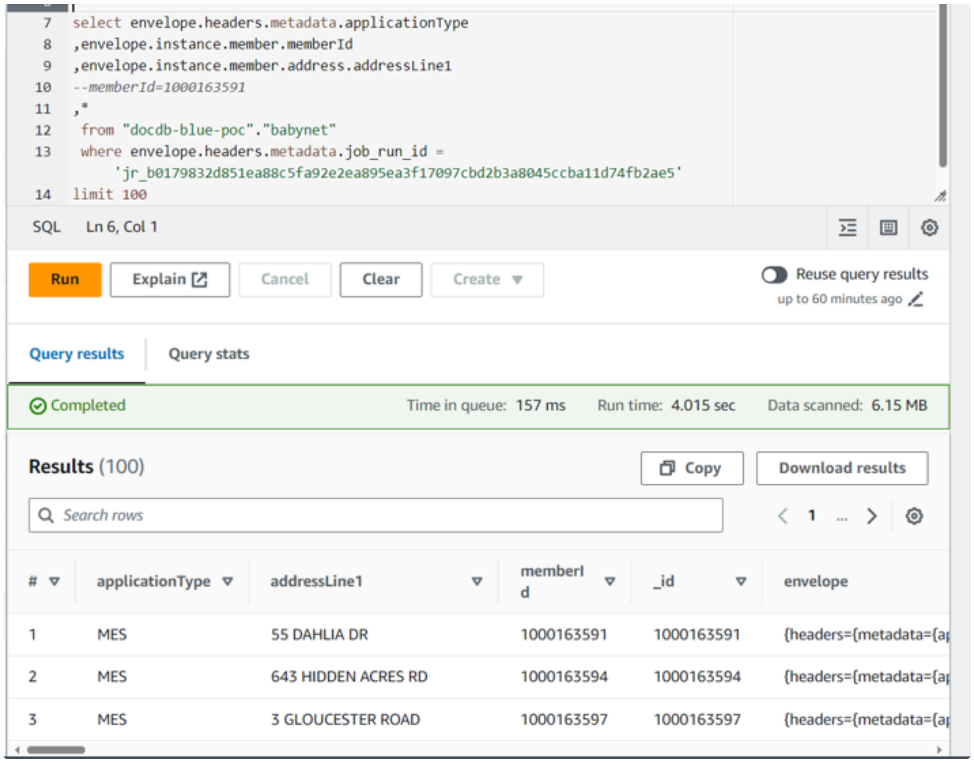
**Cloud Formation Stack** (created by AWS automatically during the data source setup)**:** **serverlessrepo-AthenaDocumentDBConnector**

1. Go to Athena through AWS console.
2. Point the data source to BlueDocumentDB.
3. Choose the required database and all the tables/collections will show up.

4. Sample count query
    a. 


5. Query for fetching a nested element and filtering the data.
    a.

```
select envelope.headers.metadata.applicationType
,envelope.instance.member.memberId
,envelope.instance.member.address.addressLine1
,*
 from "docdb-blue-poc"."babynet"
 where envelope.headers.metadata.job_run_id =
'jr_b0179832d851ea88c5fa92e2ea895ea3f17097cbd2b3a8045ccba11d74fb2ae5'
limit 100
```


select envelope.headers.metadata.applicationType
,envelope.instance.member.memberId
,envelope.instance.member.address.addressLine1
--memberId=1000163591
,*
from "docdb-blue-poc"."babynet"
where envelope.headers.metadata.job_run_id =
'jr_b0179832d851ea88c5fa92e2ea895ea3f17097cbd2b3a8045ccba11d74fb2ae5'
limit 100
SQL Ln 6, Col 1
Run
Explain
Cancel
Clear
Create
Reuse query results
up to 60 minutes ago
Query results
Query stats
Completed
Time in queue: 157 ms
Run time: 4.015 sec
Data scanned: 6.15 MB
Results (100)
Copy
Download results
Search rows
1 ...
# applicationType addressLine1 memberId _id envelope
1 MES 55 DAHLIA DR 1000163591 1000163591 {headers={metadata={ap
2 MES 643 HIDDEN ACRES RD 1000163594 1000163594 {headers={metadata={ap
3 MES 3 GLOUCESTER ROAD 1000163597 1000163597 {headers={metadata={ap